

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No-	Team No-	Project Title-
7	3	Exam Preparation App with Adaptive Quizzes

1. Abstract

The Exam Preparation App with Adaptive Quizzes is an Android-based application designed to provide personalized, efficient, and reliable exam preparation. Traditional exam preparation applications generally provide the same questions to all learners without considering their individual knowledge level and performance. To overcome this limitation, the proposed application uses the Item Response Theory (IRT) algorithm to dynamically adjust question difficulty according to each learner's responses and estimated ability.

The application follows a layered backend architecture that manages request handling, validation, business logic, data access, persistence, and response generation. It provides Question Bank Management for organizing questions based on subjects, topics, and difficulty levels, while the Adaptive Quiz Engine selects suitable questions according to learner performance. The system uses entity-relationship modelling and normalized database design to organize learner, question, quiz, result, topic, and performance data while reducing redundancy and update anomalies. SQL-based database operations are used for inserting, updating, deleting, retrieving, joining, grouping, and analyzing relational data.

The application maintains Performance Analytics to track scores, accuracy, progress, and learning patterns. Weak Topic Detection identifies topics where learners require additional practice, while the Mock Test Scheduler enables learners to plan and attempt examinations. Peer Ranking provides comparative performance information among learners. Database transactions are designed using ACID principles to maintain consistency and reliability when updating quiz attempts, scores, and learner progress, while appropriate transaction boundaries help preserve important data relationships during concurrent operations.

To improve application performance, the system uses database indexing and query optimization techniques for frequently accessed question, result, and performance data. Suitable indexes and query execution strategies help reduce unnecessary data scanning and improve response time. The application uses Kotlin for Android, Room Database for local persistence, and Firebase for cloud-based storage, synchronization, and distributed data access. Replication and scalable data access concepts are considered to support reliable access to learner data across multiple devices. Overall, the system combines adaptive learning, structured database management, transaction processing, query optimization, and distributed data management to provide a personalized, scalable, and effective examination preparation platform.

Team Details

Roll Number(s)	Name of the Student	Signature
2520030117	Konda Vaishali	Vaishali
2520030077	P. Bindhu Madhavi	Bindhu
2520030493	Alaparathi Nava Harsha	Harsha

1. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

- **Android Studio** – Development environment for the Android application.
- **Kotlin** – Primary programming language for Android development.
- **Android SDK** – For developing Android application components and interfaces.
- **Jetpack Compose / XML** – For designing the application user interface.
- **Room Database** – For local storage and management of user, question, quiz, and performance data.
- **Firebase** – For cloud storage, authentication, synchronization, and backend services.
- **SQL** – For relational database operations, queries, joins, aggregation, and data management.
- **Item Response Theory (IRT)** – For adaptive question selection and difficulty adjustment.
- **Git & GitHub** – For source-code management and version control.
- **Postman** – For testing and validating backend/API requests.

Guide Name and Signature (remarks if any) _____